2014 IEEE 38th Annual International Computers, Software and Applications Conference

The Issues of Solving Staffing and Scheduling Problems in Software Development Projects

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***Abstract*—Search-Based Software Engineering (SBSE) applies search-based optimization techniques in order to solve complex Software Engineering problems. In the recent years there has been a dramatic increase in the number of SBSE applications in areas such as Software Test, Requirements Engineering, and Project Planning. Our focus is on the analysis of the literature in Project Planning, specifically the researches conducted in software project scheduling and resource allocation. SBSE project scheduling and resource allocation solutions basically use optimization algorithms. Considering the results of a previous Systematic Literature Review, in this work, we analyze the issues of adopting these optimization algorithms in what is considered typical settings found in software development organizations. We found few evidence signaling that the expectations of software development organizations are being attended.**

Keywords*—Search-Based Software Engineering; Resource allocation; Staffing; Scheduling; Project management; Literature review

I. Introduction

In a highly competitive market, software companies must create efficient project plans and reduce development costs in order to succeed [1], [2]. The large number of dimensions and market constraints involved in software development makes Software Project Management a complex task. Software project usually involves managing large teams in a dynamic environment with unstable parameters [3]. The combination of intrinsic problem complexity and market constraints justifies the adoption of computer aided tools and practices to properly support the decision making [4]. There are many well known software project management tools, in the application sense, such as MS Project [5], Gnome Planner [6] and OpenProj [7], and, in the conceptual sense, such as COCOMO-II [8] and PMBOK [9]. Despite all these supporting tools, the project managers are responsible for producing a suitable project schedule, and they still must rely on their knowledge, experience and intuition.

In this context, Search-Based Software Engineering (SBSE) has emerged using search-based optimization techniques in complex Software Engineering problems [10]. SBSE has been applied to a diversity of Software Engineering problems [10] and, in this study, our focus is on the project planning ones. Specifically, we considered problems related to the Resource Allocation and Task Scheduling Problem (RSP), which, basically, consist of developer-to-activity allocation and the decision of when each activity should be carried out. The search algorithms used in the SBSE field are from three main groups. The first group, *Exact optimization* methods, includes classical techniques from the Operations Research area such as Branch and Bound algorithm, and Integer Linear Programming. These methods guarantee finding an optimal solution [11]. The second group, *Heuristic* algorithms, does no guarantee to find an optimal solution [11]. These algorithms search for a ‘good’ or near-optimal solution, for example, Greedy algorithms. The last group includes the *Metaheuristics* which are heuristics approaches that continue the search beyond the first encountered local optimum, for example, Genetic Algorithms (GA), Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO). The *metaheuristics* are the most preferred approach in the SBSE field [12], [13]. There are also hybrid approaches that combine techniques from these three main groups.

In the literature, several works have been done addressing the RSP [3], [4], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26]. In these studies, metaheuristics are the most applied optimization method, and the GA-based is the most used algorithm [27]. Most of these studies provided evidence of the benefits of using the optimization approaches. The improvements reported in the literature included the creation of complex plans addressing a set of restrictions as well as the development of algorithms with an enhanced performance. However only few articles mentioned the adoption of their approaches. Furthermore only few articles considered the expectations of typical software development organizations [27]. Due to the wide range of approaches and the lack of researches reporting a practical application, there is a need to investigate the aspects that should be incorporated in the RSP solutions in order to bring them closer to the industrial needs. We analyzed 52 studies addressing the RSP with the aim to answer the following research question (RQ):

RQ: What are the main issues related to the industrial adoption of RSP solutions?

The main goal of this work is to extend and improve the discussion regarding the adoption of RSP solutions in software development organizations. We discuss some of the aspects of complex project management schemes represented in the proposed models. We also discuss how the Software Engineering perspectives and stakeholders' needs were addressed.

0730-3157/14 $31.00 © 2014 IEEE
DOI 10.1109/COMPSAC.2014.96

Therefore, this paper represents an extension of our previous work [27]. Based on the results retrieved by a Systematic Literature Review (SLR), we collected new data, as described in Section III, and we present and analyze new results. We also analyzed the literature in order to identify important factors related to the project planning elaboration. We considered concepts represented in System Dynamics models featuring realistic systems [28]. The comparison between the SLR results and System Dynamics concepts helps in contrasting and analyzing the findings.

This paper is organized as follows. In Section II, we provide the background relevant to the research, and we set forth the results of our previous SLR. Section III defines the research methodology. In Section IV, we present the results, while Section V outlines its validity threats. The paper is concluded in Section VI.

II. Background

In this section, we introduce the concepts applied for our research analysis. We also briefly describe our previous literature review.

A. Adoption

Venkatesh and others [29] and Sabherwal and others [30] have examined the individual's adoption and continuance usage intentions of Information Technology. There is a consensus that adoption is impacted by the attributes of innovation, the individual characteristics, and other contextual factors [29], [30], [31].

In this work we focused on the study of attributes of a theoretical adoption model [32]. This theoretical model incorporated varying influences of different factors over time. The innovation attributes [31], [33] - described as perceived usefulness, perceived compatibility, and perceived ease of use - and individual characteristics [31], [34] - described as expertise, personal innovativeness, and self-efficacy - drive adoption in the early stage, whereas the innovation attributes and contextual factors [29], [31], [35] - described as facilitating conditions and social influence - drive adoption and continuance of usage in the later stage. In our study, we were able to collect some evidence from the literature related to the innovation attributes. The analysis of these attributes guided us on answering our research question.

According to Rogers [33] the adoption is determined by several innovation attributes. Three important attributes that we apply in our analysis are [31], [36]: perceived usefulness, perceived ease of use, and work compatibility. *Perceived usefulness* refers to the perceived benefits of technology adoption. *Perceived ease of use* refers to how easily a technology is adopted. *Work compatibility* describes the fit between the technology adoption and the work environment. All these attributes are expected to produce a positive influence on the project managers' intention to adopt a new technology [31].

B. Related Work

To the best of our knowledge only a recent survey has been published on Search-Based Software Project Management (SBSPM) with the aim to present an overview of this area [37]. This survey discusses four SBSPM problems: staffing, scheduling, risk and effort estimation. According to them, promising areas for investigation in SBSPM include:

1) Interactive optimization: This technique explores project manager's expertise for fitness computation. Since the project management task is inherently human-centric, it would be natural to explore the use of this technique.
2) Dynamic Adaptive Optimization: Solutions can be computed in real time and the decision makers can continually interact with the optimization, exploring the implications of their decision. The optimization provides best-so-far solutions as the decision maker interacts.
3) Multi-Objective Optimization: Recent SBSE work has followed a more multi-objective style of approach. A trend in SBSPM is to focus on decision support in complex multi-objective problem spaces.
4) Co-Evolution: Cooperative co-evolutionary model is a natural model for project management and it deserves more attention in the SBSPM field.
5) Software Project Benchmarking: One of the core problems in the SBSPM is the lack of available real world project data. Researchers are making some effort to employ real data and to deal with more human-centric problems.
6) Confident Estimates: Considering the difficulties in obtaining management processes exact estimates, it would be beneficial to introduce some level of uncertainty and measure their effects.
7) Decision Support Tools: In order to bring successfully SBSPM methods into practice, it is needed to develop decision support tools. These tools will allow an extensive evaluation of the interface considering technical aspects and other related socio-technical issues.

Besides many advances in the SBSPM field, this survey confirms our findings: (i) missing SBSPM tool support; (ii) not realistic representation of central attributes, such as skill; and (iii) lack of real data and corresponding evaluations.

C. Project factors

Many different types of models have been developed to address project management features and characteristics. System Dynamics is one way of modeling features found in realistic systems [28]. Modeling important components of actual projects relates directly to the experiences of practicing managers. In software projects, these include development processes, resources, managerial mental models, and decision making. According to Lyneis and Ford [28], the development of project models has captured the majority of the important features of realistic development projects. Therefore we decided to collect the most important factors from these models and contrast them with the proposed SBSPM solutions.

While specific models differ in their level of details, their formulations have been extensively documented. In these models, software projects almost always consist of a collection of tasks. These tasks can be performed serially or in parallel, they may represent the entire project, or may represent development phases (e.g., requirements specification, coding or testing) [28]. These tasks are interrelated, thus productivity can vary considerably [38]. This variation depends on many factors such as allocation fraction [39], schedule pressure [38], learning speed [38], experience [28], communication difficulties [40], [41], morale [42], and overtime work [41]. A number of other concepts were also represented such as project deadline, error rate, tasks effort, and resources cost. Modelers improved project models by adding human features [28]. These features reflect factors that amplify the negative or positive effects on project development.

We considered part of the factors for describing the RSP, including the human perspective. These factors brought the problem solutions closer to what happens in software development organizations.

D. Overview of the previous work

In our previous work, 52 primary studies were obtained from an SLR [27]. An SLR is a means of evaluating and interpreting available research relevant to a particular research question [43]. Here, we present the main results achieved with this review.

1) We have extracted text containing information describing the concepts used in the problem model. We observed that the most common concepts represented in the models are: resource, project, task and skill (see Table I). These concepts are specialized as the context is more detailed. For example, (i) when there is outsourced work, we have resources within the company and outside the company. External resources may have restrictions related to attending trainings and, in consequence, improving their skills; (ii) developments that involve companies around the world (*Global Software Development*) define teams working in different time zones; (iii) in a portfolio planning, it is interesting to know the weight and priority of each project.
2) Important aspects were not addressed in these research papers. For example, only 10 studies (19%) considered different employees' productivities and, even intuitively, the project manager considered this attribute during planning. Five studies (10%) defined that the skill level of a resource can change during the life of the project, such as the engineers who gain experience by previous activities. Only one study includes communication overhead factor when the team size increases. Process change and technology adoption, two aspects that constantly happen in a project development, are usually omitted. This lack of investigation encourages further research in the SBSPM field.
3) The most applied optimization technique is the evolutionary (56%, 29 studies), and the GA-based are the most used in the researches (33%, 17 studies). Heuristic techniques, such as Greedy algorithms, are applied by 8 studies (15%). Other techniques identified included: (i) other metaheuristics techniques (21%, 11 studies), such as PSO, ACO and Simulated Annealing, (ii) exact methods, including Branch and Bound and Integer Linear Programming (6%, 3 studies) and (iii) Hybrid approaches (17%, 9 studies). A large percentage of studies 88% (46 studies) addresses single-objective approaches. Considering the multi-objective evolutionary technique, NSGA-II (10%, 5 studies) is the most used MOEA (Multi-objective Evolutionary Algorithm), followed by SPEA2 (8%, 4 studies) and MOCell (6%, 3 studies).
4) As confirmed by other researches [13], [44], [45], there is a trend to use new algorithms and multi-objective approaches. The first MOEA study addressing RSP was published in 2009. These techniques are more suitable to deal with complex Software Engineering problems than other classical techniques.
5) All studies stated clearly the optimization goal and the concepts used in the conceptual model (see Table II). However, most of the studies (62%, 32 studies) did not explicitly define the context where the approaches can be applied. In addition, 35 studies (67%) did not present the proposed solution (e.g., pseudo-code) and only 14 articles mentioned the threats to validity. Most of the validations (42 studies, 81%) were carried out without a proper statistical analysis (descriptive or hypotheses testing). Without this kind of analysis, it is difficult to infer that one approach perform better than other one, as well as, to interpret or replicate the results.
6) Although 17 (33%) studies use real data instances, we did not identify a conclusive evidence of the practical application of the SBSPM algorithms. Only two articles, [24] and [46], mentioned the use of the algorithms in a software development organization. Therefore, it is needed more investments to bridge the gap between the research and practical field.
7) Most instances used in the validation step are not impressive in size. When analyzing the instances' sizes, specifically, the numbers of 'Tasks' and 'Resources', we concluded that most of the instances are small or medium. Random instances or fictitious project data were used by 35 studies (67%). Real instances were used in 17 (33%) studies.
8) The comparison among different approaches is the most common evaluation strategy (56%, 29 studies), followed by the analysis of different parameters values (37%, 19 studies). Although the comparisons are the most used approaches, the validity of the results, in most of the cases, were not strength with application of statistical techniques, such as hypothesis testing or correlation. The impact of this lack of statistical analysis is increased by unsatisfactory evaluation of threats to validity.

A detailed discussion of these results can be found in Peixoto and others [27].

III. Methodology

For our work, we considered the 52 primary studies retrieved by the SLR. Our aim is to identify aspects related to the difficulties in adopting the proposed RSP solutions by software development organizations. In general, these primary studies are structured in three parts: (i) a description of a mathematical problem formulation containing variables, data, constraints and

TABLE I. RSP CONCEPTS [27].

Concept	Description	Common attributes	Frequency
Task	Task is an element of project work decomposition. Although task and activity definitions are different, we grouped this information in order to ease our analysis.	*Task identification*, *Precedence* (in most studies represented as a Task Precedence Graph - TPG), *Skill requested* (level and the skill set), *Effort* (in many studies estimated using COCOMO [8] or some Fuzzy logic), *Maximum headcount* (maximum number of employees who can be assigned to the task), *Complexity*, *Priority*, and *Duration*.	48
Resource	Employee who performs a task.	*Resource identification*, *Role* (employees' role in the software process), *Executable task type* (tasks that the resource are allowed to execute), *Salary*, *Availability* (start and end date), *Productivity*, *Skills* (level and the skill set), *Experience* (experience in some task or period of time that the employee carried out a specific task), *Dedication*, *Maximum daily workable hours*, *Learning speed*, and *Overwork* (amount allowed and the cost).	43
Project	A software project which has a defined start and end date.	*Duration*, *Budget*, *Benefit rate* (when a project is finished before its deadline), *Penalty rate* (when a project is late), *Cost*, *Weight* (organization's priority).	20
Skill	The employees' expertise.	*Skill identification*, and *Level* (skill categories).	14
Team	Group of employees responsible for doing the task.	*Size*, *Communication overhead* (Team size dependent measure), and *Specialization*.	13
Artifact	A product produced during software development.	*Size*, *Quality* (number of defects), and *Dependencies*.	10
Other concepts	Additional concepts that do not appear very often in the studies. Examples include: Process-related concepts (Phase, Increment, Iteration, Risk, and Release) and Business model concepts.	*Phase, Increment penalty* (when a project is late), and *Sites* (number of different organizations in a GSD).	27

TABLE II. OPTIMIZATION' GOALS [27].

Goal	Description	Frequency
Duration	The length of time that software product takes to be available to customers.	44
Cost	The monetary resources needed to complete the software development project.	25
Skill	The sum of skill levels for the project.	5
Productivity	Technical efficiency of the development team.	4
Quality	Number of defects detected after delivery.	4
Other goals	Examples: *Utility* (weights the importance of keeping the rescheduled project similar to the initial plan), *Stability* (weights the importance of responding to the disruption, e.g., requirements changing), *Resource usage* (number of developers assigned to each task), and *Overwork* (when the actual dedication is above the maximum dedication).	9

objectives; (ii) a brief description of the optimization algorithms; and (iii) a description of the validation approaches and outcomes. Drawing an analogy with the software development processes, we can relate this structure to the following stages of software development: problem conception, development and validation. We collected data related to these three stages in order to support our claims and answer our RQ.

1) Problem conception: We collected, from the primary studies, additional information concerning the concepts and their attributes used in the mathematical formulation. These concepts map the aspects considered by project managers during the scheduling and resource allocation task into an abstract representation. These identified concepts gave us a general conceptual model described in the next section. This conceptual model coding clusters related case studies' data into smaller number of sets, allowing a more integrated schema for analyzing the results. In addition, we searched for information that identifies who participated in these conceptual representation tasks. When we considered the innovation attributes (as listed in Table III), the project concepts and the stakeholder's involvement information were key factors for the verification of the 'work compatibility' attribute.
2) Development: Since our goal is to evaluate technology adoption issues, we focused more on Software Engineering aspects than on implementation issues of the optimization techniques. In this stage, we collected information regarding the usage of any Graphical User Interface (GUI) and possible integration with other project management tools, for instance, with MS Project. These two aspects are important for the verification of the 'work compatibility' and 'perceived ease of use' attributes.
3) Validation: Specific validation techniques, such as experiments, are valuable tools for all software engineers who are involved in evaluating and selecting different methods, techniques, languages and tools [47]. We identified which stakeholders were involved in the validation step, the methodologies and data used to validate the optimization approach, and the benefits achieved by the use of the proposed approach. This information is important to define the 'work compatibility' and 'usefulness'.

TABLE III. MAPPING ATTRIBUTES.

Innovation attribute	Information collected
Work compatibility	Project concepts, integration with other tools, graphical user interface, and stakeholders' involvement in the problem definition and in validation.
Usefulness	Benefits, and stakeholders' involvement in validation.
Perceived ease of use	Integration with other tools, and graphical user interface.

Table III lists the mapping between the information collected and the innovation attributes.

For our data analysis, we have extracted text containing information presented in Table III from the primary studies. The evidence developed in this research rests mainly on the identification and analysis of this information. We also triangulated the results with some practitioners driven features collected from System Dynamics models (see Figure 1). This triangulation helped establish the importance of the modeling concepts and perceived relationship between these concepts and the research aims. Reliability and validity are here dependent on the careful analysis of the research papers.

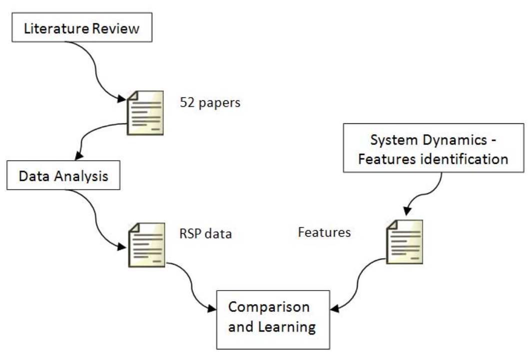

Fig. 1. Research design.

IV. RESULTS

In this section we describe our empirical findings.

A. Problem conception

In our previous work [27], we have extracted text containing information describing the concepts used in the problem model. Table I lists the main concepts. As we described above, four important concepts are 'Task', 'Resource', 'Project' and 'Skill'.

Figure 2 depicts a UML-based [48] model representing these concepts. The goal of this model is to display the main concepts with their most common attributes. This figure was derived from the primary studies and it consists of a generalization of the usual elements found in the primary studies' problem model. It is by no means a comprehensive model.

Table II shows that 'duration' and 'cost' are the most measured goals. The change in these goals determines an effect similar to the relationship of cost, time and scope, in the 'Project Management Triangle'. In this triangle, the relationship is such that if any one of the factors changes then at least one of the others will be affected. Moreover, it is also not possible to improve all factors at the same time [9].

In the analyzed papers, there is a clear focus on the basic concepts and their attributes. When we contrast with other models (e.g., System Dynamics models) it seems that important aspects were not addressed, such as productivity, learning speed, and communication overhead. One possible cause is the lack of the main stakeholders' participation during the model definition. Indeed, we were not able to identify who (besides the authors) participated in most of the conceptual model elaboration. One exception is the work of Kang and others [49]. In this work, the constraints were identified based on the literature and interview with experts in the industry. They proposed individual-level constraints, e.g., allocating developers to the same modules in each development phase, and team-level constraints, e.g., discouraging the share of developers between teams. As far as we could observe, the SBSPM researches, in particular the ones related to the RSP, have room for improvement regarding the investigation of technology, process, human, and organizational factors that affect project development.

Our first observation is:

In order to represent a model closer to the industrial environment more factors such as, 'human', 'organizational' and 'process' need to be taken into account.

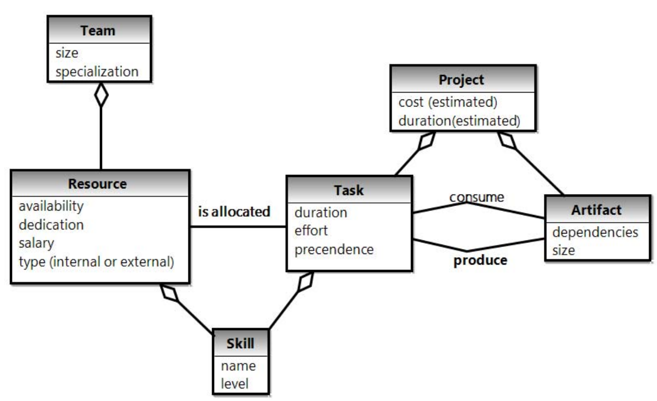

Fig. 2. Resource allocation and Scheduling Problem model.

B. Development

Compared with other project management tools and methodologies, the scheduling and resource allocation solutions are especially complex to be directly adopted by project managers without some tool support. Mainly because it involves knowledge covered by other research area: Operational Research. Therefore we believed that two aspects will ease project managers' life: integration with tools they are acquainted with and a GUI for data input and output. We analyzed the 52 research papers in order to investigate these aspects.

Considering the first aspect, we expected to find descriptions about the integration with project managers commonly used tools. However, we found only one article describing the possibilities to integrate with other tools and environments [23]. In this case, a specific tool imports entry information and exports results in a XML based file format. In addition, we found examples of integration with other approaches with the aim to study the solution effectiveness, such as in the work of Xiao and others [50].

Considering the GUI, we identified four articles [23], [51], [52], [53] describing its development and usage. Stylianou and others [51] designed and implemented a prototype tool for software project managers that integrates several optimization techniques for scheduling and staffing. The prototype results are displayed in the form of tables and charts with which a software project manager is able to compare solutions and determine which staffing strategy is best to adopt. Li and others [53] showed a prototype that integrates scheduling into software release planning. Given an expected release date, it is possible to select the requirements for the next release and calculate an on-time delivery project plan. Barreto and others [23] showed a tool that allows the manager to choose the productivity model for each project and to setup model parameters. In addition, it allows to maintaining information describing the professionals and the activities. Finally, Sanal [52] only briefly mentioned the existence of an interface. As far as we could observe, the SBSPM researches, in particular the ones related to the RSP, have room for improvement regarding GUI development and tools integration.

Our second observation is:

In order to be compatible to the industrial environment and to be easy to use, there is a need for more research on GUI development and tools integration.

C. Validation

We had categorized the studies according to the methodology used to validate the proposed approach [27]. Since most of the studies did not mention explicitly this information, we collected it from reading the papers. The following categories were identified in the selected papers:

- **Quasi-experiment** if the study validates the research by measuring the effects on one variable when another is manipulated and the subjects are not randomly assigned to treatments [54], [55]. When the subjects evaluate the approach without rigorous control, we classified the study as an **informal experiment**.
- **Sensitivity analysis** if the study investigates the sensitivity of the optimization approach results, changing the input values [12]. In this case, it is essential to know the most important impacts of the solution's factors.
- **Comparative analysis** if it is collected quantitative and/or qualitative data when comparing the proposed approach with other optimization approach. This analysis can employ statistical methods to compare the results.
- **Report** if there is only a description of the algorithm application.

Most of the studies employed Comparative analysis (46%, 24 studies), followed by Sensitivity analysis (31%, 16 studies), Report (15%, 8 studies) and Experiment (8%, 4 studies). Considering the experiment, only one study applied a 'quasi-experiment' approach [23] and three 'informal experiments' [3], [56], [57]. Although the 'comparative analysis' was the most used approach, the validity of the results, in most of the cases, did not include suitable statistical techniques, such as hypothesis testing or correlation. Most of the 'comparative analysis' studies did not mention the use of a baseline. When a baseline was defined, the random search was the most common method. Since our intention is to investigate whether the main stakeholders[1] validated the approach, we focused, below, on the discussion of the experiments.

With the aim to generalize from an experiment with a given group of subjects, the information about the sample characteristics are a central issue [58]. Since the Software Engineering community does not know which variables are important as background in an experiment [59], we collected the variables reported in the analyzed research papers. In these four papers (one containing a 'quasi-experiment' and three 'informal experiments'), the level of detail describing the experiments varies substantially. Table IV shows the characterization.

Overall, the experiments have a similar goal: compare the results from the automated approach against the project managers' intuition. They performed a qualitative analysis in order to get an indication whether the use of the proposed approach might be useful for a project manager who needs to schedule and staff a software project. The project instances used 15, 7, 5 and 3 resources and 25 and 7 tasks, which we considered small. The four experiments employed people with professional experience. In the researchers' viewpoint, the proposed approaches provided solutions that are better than the ones provided by the targeted stakeholders.

Our third observation is:

In order to be adopted in software development organizations, more conclusive empirical evidence needs to be reached. Mainly conducting controlled experiments, with projects' complexity and size closer to the ones in the industry.

D. Discussion

What are the main issues related to the industrial adoption of RSP solutions? We had collected in our previous research and refined in this work some findings (F) related to the difficulties of adoption: (F1) restricted number of represented concepts; (F2) randomly generated instances, not reflecting real data; (F3) small instances sizes used in validations;

[1]The main or targeted stakeholders are the project managers who have to deal with operational aspects of project scheduling and resource allocation.

TABLE IV. EXPERIMENTS DETAILS.

Categories	Chang and others [3]	Barreto and others [23]	Maia and others [56]	Britto and others [57]
Subjects	2	16	3	5
Experience	Senior project managers	M.Sc. and D.Sc. students, with software development experience in both academic and industrial projects.	Project managers.	Software engineer. Some of them having software test experience.
# Sessions	1	4	1	1
Duration (per section)	3 hours	1 hour	4 hours	Not specified.
Task	Assign the employees to tasks. Project with 25 tasks, 15 resources, and 12 types of skills.	Select the cheapest team to perform a fictitious project by using *ad hoc* procedures. Project with 7 tasks, 7 resources, and 10 types of skills.	Allocate teams with 3, 5 and 7 developers trying to balance between the team cost and team productivity.	Not specified.

(F4) restricted participation of the main stakeholders; (F5) insufficient investments in GUIs; (F6) lack of integration with other tools and environments. We also identified a finding that would ease the adoption: (F7) RSP solutions outperformed the expert' assignments. We emphasize the following differences between our previous work [27] and this work. Findings F1, F2 and F3 were reported in our previous work. Findings F4, F5, F6 and F7 are specific of the present work. Below we discuss each of these findings.

F1: The immaturity of organizational development processes is one of the reasons that explains the difficulties of getting software measurements. Unfortunately, there is no widely accepted measurement framework. However, project managers, even without a defined measurement process or approach, use a certain informal classification scheme. At least part of these classification schemes may be incorporated in the conceptual models.

We can triangulate our interpretations with what other researchers use in their models. When we analyze the System Dynamics models, we can observe some important features that are used to represent the dynamism of a software development project, such as human factors, ripple effects (unintended side effects to explain project behavior and performance), and rework cycle [28]. Features associated to developers characteristics and behavior are central to System Dynamics models. They could be better represented in the search-based models. This would improve the ability to represent and generate solutions of more realistic projects.

In addition, Software Engineering 'rules' usually employed in supporting tools as simulations, can also be helpful when generating the model and the instances. For example, two 'rules' that can be mapped in the problem model formulation are: '*Two factors that affect productivity are work force experience level and level of project familiarity due to learning-curve effects*' [60] and '*The training of new employees is usually done by the 'old-timers', which results in a reduced level of productivity on the 'old-timers' part. Specifically, on the average, each new employee consumes in training overhead 20% of an experienced employee's time for the duration of the training or assimilation period*' [60]. An extensive list of Software Engineering 'rules' can be found in Navarro [61] and Kupsch [62].

F2: Most instances are randomly generated, not reflecting real data. The considerable effort needed to collect data is a de-motivator factor for the elaboration of real instances. In the Software Engineering area, it is difficult to find enough data in the literature or with appropriate quality level. Even when available, real-world data sets often contain noisy, irrelevant or redundant variables [63]. Beyond the organizational measures, we can appeal, for example, to specialists. However, we still need to verify whether the results approximate the real world and whether they are suitable for the organizational environment.

F3: Most instances do not reflect the number of tasks and resources in a real project planning. In order to check the evaluation approaches, we aggregated the number of tasks and resources input data, i.e. the instances' sizes, in the following sets: small - up to 30 tasks and 5 resources; medium- between 30 and 90 tasks and between 5 and 15 resources; large - above 90 tasks and 15 resources. A study has to attend the limits of both concepts, for example a study with 100 tasks and 10 resources is considered attending a medium set.

Most of the instance sizes are small (62%, 32 studies), followed by medium (29%, 15 studies) and large (9%, 5 studies) instances. Therefore, wrong or imprecise conclusions might be drawn when only considering them. Analyzing the causes of this limitation, we returned to the discussion of data collection in Software Engineering field and their palliatives solutions, such as having the help of a specialist.

F4: One success factor of a Software Engineering research is the adoption of its results into software development organizations. Successful Empirical Software Engineering research will have its results transferred to industrial applications [64]. One main problem that we observed in the four articles, that reported an experiment, is the lack of realism. The experiments would be more realistic if they are run on real instances with bigger sizes, with the participation of a larger number of stakeholders, in particular, project managers, and in a more controlled way.

A common problem of the SBSPM research papers is the lack of empirical validation with stakeholders. This problem threatens our evaluation results. Therefore, more experiments should be carried out in order to identify cause-effect relationships, which will influence industrial relevance.

F5 and F6: Since only few investments were made in GUI development and tool integration, we cannot affirm that there is a trend among the SBSPM research articles to provide support to project managers. Indeed, only two studies have mentioned a practical application of their approach (i.e. in industry). One main reason that may justify the investments is the lack of optimization knowledge of project managers. The GUI and the integration with other tools would be an important help for the project managers regarding this optimization knowledge.

F7: The SBSPM techniques have potential to provide decision support for software project managers. The SBSPM researches showed that the solutions to the RSP are viable, even if the models do not reflect the complex dynamics of software projects. In general the solutions outperformed the experts' assignments and other specific approaches, for example, standard solvers such as CPLEX.

When we evaluated the innovation attributes, we concluded that almost all research papers do not address suitably the issues related to 'work compatibility', 'perceived ease of use' and 'usefulness'. These issues encourage further research in the SBSPM field.

V. Validity Threats

The use of systematic procedures itself helps to avoid problems with the selection and analysis of the studies. Even though this study has been supported by a pre-defined study protocol and conducted in a systematic way and under the guidance of experts, it has some limitations. Below we briefly discuss these limitations considering four aspects [65].

Construct Validity is concerned to what extent the selected studies represent what we aim to investigate [47]. One possible threat to construct validity is exclusion of relevant studies. Since the search for primary studies is based on the search string, each SLR is likely to miss relevant studies if this string is not properly chosen. In order to minimize this threat, we conducted a rigorous evaluation of our search strategy [27].

Reliability refers to the possibility of repetition of data collection and analysis, achieving the same results, by other researchers [65]. Some countermeasures were taken to reduce the threats to Reliability. Since we followed Kitchenham's [65] procedures (i.e. we defined a research question, the selection process, inclusion/exclusion criteria and quality criteria), we believe that we detailed the protocol enough to provide an assessment of how we achieved the final set of papers and how we conducted the analysis. We assessed the consistency of data extraction, by extracting for the second time a set of selected primary studies. In addition, the first author participated in all the steps and review sections, either as a reviewer or as a supporter. The participation of the first author ensured a series of countermeasures related to possible threats. We expect that, since we reported all the procedures taken and limitations, this research has high reliability and can be replicated by other researchers.

Internal Validity is concerned to what extent the design and conduction of the study is likely to prevent systematic errors [65]. *Reviewer bias* is a potential threat to Internal Validity as the extracted data have a qualitative nature. In order to mitigate this threat, the concepts and their attributes were grouped in order to facilitate the studies classification and comparison.

External Validity is concerned with the generalization of results outside the scope of the study [47]. The results of this review were considered with respect to specific studies in the search-based domain. Therefore, the conclusions and classifications were valid only in this given context. The results of our current study were drawn from qualitative analysis and can serve as a starting point for future researches. Additional studies can be analyzed accordingly.

VI. Conclusion

This paper extended and improved the discussion of a previous SLR about the search-based optimization solutions of the resource allocation and scheduling problem. Our goal was to present some reasons that would explain the adoption or non-adoption in industrial settings of the considered optimization solutions.

In order to assist in the identification of the reasons, we used three innovation attributes: 'work compatibility', 'usefulness', and 'perceived ease of use'. These innovation attributes worked as a guidance to help in the definition of the information to be collected. The identification of this information was eased by the fact that the papers followed traditional mathematical modeling. First, there is a description of the problem model, i.e. a description of the problem formulation containing variables, data, constraints and objectives. Second, there is a brief description of the optimization algorithms. Finally, there is a description of the validation approaches and outcomes.

After analyzing 52 articles, we observed that it is needed more investments to bridge the gap between the RSP research and practical Software Engineering field. Besides the benefits of the optimization approaches, we concluded that there are lots of opportunities for improvement in the SBSPM area. There is a restricted number of represented concepts in the conceptual models. We confirmed this evidence after triangulating the modeled concepts with the System Dynamics ones. We did not find a suitable investigation regarding human, organizational, technology and process factors that affect a project development. Regarding the instances, the researchers extensively used randomly generated instances for the approach validation, not reflecting real data. In addition, these instances are not impressive in size. In practice, only few articles mentioned the effective adoption of the proposed approach in environments where they were justifiable. We considered that the targeted stakeholders should be more involved during the problem analysis and solution. Finally, almost no integration facilities with other project management tools and GUI investments were reported in these studies.

The results encourage further research in order to better bridge research and practice. One future work consists in the investigation of the main aspects that project managers consider during the planning task. Project managers often have different views and backgrounds which may prioritize certain factors over others, with further analysis, this could enrich this area of research. Another future work consists in perfecting conceptual resource and scheduling meta-models which will represent the main concepts that should be addressed by the optimization algorithms. Finally, we believe that some guidance for results evaluation and communication will improve the works' analyses quality.

Acknowledgment

This work was supported by CAPES, CNPq and FAPEMIG, entities of the Brazilian government.

References

[1] N. Nan and D. Harter, "Impact of Budget and Schedule Pressure on Software Development Cycle Time and Effort," *IEEE Transactions on Software Engineering*, vol. 35, no. 5, pp. 624–637, 2009.

[2] W.-N. Chen and J. Zhang, "Ant Colony Optimization for Software Project Scheduling and Staffing with an Event-Based Scheduler," *IEEE Transactions on Software Engineering*, vol. 39, no. 1, pp. 1–17, 2013.

[3] C. K. Chang, H.-y. Jiang, Y. Di, D. Zhu, and Y. Ge, "Time-line based model for software project scheduling with genetic algorithms," *Information and Software Technology*, vol. 50, no. 11, pp. 1142–1154, oct 2008.

[4] E. Alba and J. F. Chicano, "Software project management with GAs," *Information Sciences*, vol. 177, no. 11, pp. 2380 – 2401, 2007.

[5] M. Corporation, "Microsoft Office Project," http://office.microsoft.com/project, 2007, [Online]. Accessed 10th May 2013.

[6] G. Project, "Gnome planner," https://live.gnome.org/Planner, 2010, [Online]. Accessed 10th May 2013.

[7] OpenProj, "OpenProj - Project Management," http://sourceforge.net/projects/openproj/, 2012, [Online]. Accessed 10th May 2013.

[8] B. W. Boehm, C. Abts, A. W. Brown, S. Chulani, B. K. Clark, E. Horowitz, R. Madachy, D. J. Reifer, and B. Steece, *Software Cost Estimation with COCOMO II*, 1st ed. Upper Saddle River, NJ, USA: Prentice Hall PTR, 2000.

[9] P. M. Institute, *A Guide to the Project Management Body of Knowledge (PMBOK®Guide)*, 5th ed. Project Management Institute, 2013.

[10] M. Harman, S. A. Mansouri, and Y. Zhang, "Search-based Software Engineering: Trends, Techniques and Applications," *ACM Computing Surveys*, vol. 45, no. 1, pp. 11:1–11:61, December 2012.

[11] F. Rothlauf, *Design of Modern Heuristics: Principles and Application*. Springer Berlin Heidelberg, 2011.

[12] M. Harman, "The Current State and Future of Search Based Software Engineering," in *2007 Future of Software Engineering*, Minneapolis, MN, USA, 2007, pp. 342–357.

[13] T. E. Colanzi, S. R. Vergilio, W. K. G. Assunçao, and A. Pozo, "Search Based Software Engineering: Review and analysis of the field in Brazil," *Journal of Systems and Software*, vol. 86, no. 4, pp. 970 – 984, 2013.

[14] L. Özdamar and E. Alanya, "Uncertainty modelling in software development projects (with case study)," *Annals of Operations Research*, vol. 102, no. 1-4, pp. 157–178, 2001.

[15] C. K. Chang, M. J. Christensen, and T. Zhang, "Genetic algorithms for project management," *Annals of Software Engineering*, vol. 11, no. 1, pp. 107–139, 2001.

[16] H.-T. Tsai, H. Moskowitz, and L.-H. Lee, "Human resource selection for software development projects using Taguchi's parameter design," *European Journal of Operational Research* , vol. 151, no. 1, pp. 167 – 180, 2003.

[17] G. Antoniol, M. Di Penta, and M. Harman, "A robust search-based approach to project management in the presence of abandonment, rework, error and uncertainty," in *10th International Symposium on Software Metrics*, 2004, pp. 172–183.

[18] J. Duggan, J. Byrne, and G. Lyons, "A task allocation optimizer for software construction," *IEEE Software*, vol. 21, no. 3, pp. 76–82, 2004.

[19] O. Bellenguez and E. Néron, "Lower Bounds for the Multi-skill Project Scheduling Problem with Hierarchical Levels of Skills," in *Proceedings of the 5th International Conference on Practice and Theory of Automated Timetabling*, ser. PATAT'04, Pittsburgh, PA, 2005, pp. 229–243.

[20] G. Antoniol, M. Di Penta, and M. Harman, "Search-based techniques applied to optimization of project planning for a massive maintenance project," in *Proceedings of the 21st IEEE International Conference on Software Maintenance*, 2005, pp. 240–249.

[21] T. Hanne and S. Nickel, "A multiobjective evolutionary algorithm for scheduling and inspection planning in software development projects," *European Journal of Operational Research*, vol. 167, no. 3, pp. 663 – 678, 2005.

[22] P. Jalote and G. Jain, "Assigning tasks in a 24-h software development model ," *Journal of Systems and Software*, vol. 79, no. 7, pp. 904 – 911, 2006.

[23] A. Barreto, M. de O. Barros, and C. M. Werner, "Staffing a software project: A constraint satisfaction and optimization-based approach," *Computers & Operations Research* , vol. 35, no. 10, pp. 3073 – 3089, 2008.

[24] P. Kapur, A. Ngo-The, G. Ruhe, and A. Smith, "Optimized staffing for product releases and its application at Chartwell Technology," *Journal of Software Maintenance and Evolution: Research and Practice*, vol. 20, no. 5, pp. 365–386, 2008.

[25] S. Gueorguiev, M. Harman, and G. Antoniol, "Software Project Planning for Robustness and Completion Time in the Presence of Uncertainty Using Multi Objective Search Based Software Engineering," in *Proceedings of the 11th Annual Conference on Genetic and Evolutionary Computation*, ser. GECCO '09. Montréal, Québec, Canada: ACM, 2009, pp. 1673–1680.

[26] M. Di Penta, M. Harman, and G. Antoniol, "The use of search-based optimization techniques to schedule and staff software projects: an approach and an empirical study," *Software: Practice and Experience*, vol. 41, no. 5, pp. 495–519, 2011.

[27] D. C. C. Peixoto, G. R. Mateus, and R. F. Resende, "Evaluation of the Search-Based Optimization Techniques to Schedule and Staff Software Projects: a Systematic Literature Review," Federal University of Minas Gerais, Brazil, Technical Report, 2013, [Online]. Available: http://www.dcc.ufmg.br/∼cascini/.

[28] J. M. Lyneis and D. N. Ford, "System dynamics applied to project management: a survey, assessment, and directions for future research," *System Dynamics Review*, vol. 23, no. 2-3, pp. 157–189, 2007.

[29] V. Venkatesh, M. G. Morris, G. B. Davis, and F. D. Davis, "User Acceptance of Information Technology: Toward a Unified View," *MIS Quarterly*, vol. 27, no. 3, pp. 425–478, 2003.

[30] R. Sabherwal, A. Jeyaraj, and C. Chowa, "Information System Success: Individual and Organizational Determinants," *Management Science*, vol. 52, no. 12, pp. 1849–1864, 2006.

[31] Y. Sun, A. Bhattacherjee, and Q. Ma, "Extending technology usage to work settings: The role of perceived work compatibility in ERP implementation ," *Information & Management*, vol. 46, no. 6, pp. 351 – 356, 2009.

[32] Y. Sun and A. Jeyaraj, "Information technology adoption and continuance: A longitudinal study of individuals' behavioral intentions ," *Information & Management*, vol. 50, no. 7, pp. 457 – 465, 2013.

[33] E. M. Rogers, *Diffusion of Innovations, 5th edition*. The Free Press, New York, 2003.

[34] O. Kwon, K. Choi, and M. Kim, "User Acceptance of Context-aware Services: Self-efficacy, User Innovativeness and Perceived Sensitivity on Contextual Pressure," *Behaviour & Information Technology*, vol. 26, no. 6, pp. 483–498, Nov. 2007.

[35] F. D. Davis, "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology," *MIS Quarterly*, vol. 13, no. 3, pp. 319–340, 1989.

[36] F. D. Davis, R. P. Bagozzi, and P. R. Warshaw, "User Acceptance of Computer Technology: A Comparison of Two Theoretical Models," *Management Science*, vol. 35, no. 8, pp. 982–1003, Aug. 1989.

[37] F. Ferruci, M. Harman, and F. Sarro, "Search-based software project management," in *Software Project Management in a Changing World*, G. Ruhe and C. Wholin, Eds. Springer, 2014, p. in press.

[38] R. J. Madachy, "A Software Project Dynamics Model for Process Cost, Schedule and Risk Assessment," Ph.D. dissertation, Faculty of the Graduate School, University of Southern California, 1994.

[39] N. R. Joglekar and D. N. Ford, "Product development resource allocation with foresight," *European Journal of Operational Research*, vol. 160, no. 1, pp. 72 – 87, 2005.

[40] T. K. Abdel-Hamid, "The dynamics of software development project management: an integrative system dynamics perspective," Ph.D. dissertation, MIT Sloan School of Management, Cambridge, MA, 1984.

[41] D. N. Ford and J. D. Sterman, "Dynamic modeling of product development processes," *System Dynamics Review*, vol. 14, no. 1, pp. 31–68, 1998.

[42] A. Rodrigues, "The application of system dynamics to project management: an integrated methodology (SYDPIM)," Ph.D. dissertation, Department of Management Science, University of Strathclyde, UK, 2000.

[43] B. Kitchenham and S. Charters, "Guidelines for performing Systematic Literature Reviews in Software Engineering, version 2.3," Software Engineering Group, School of Computer Science and Mathematics, Keele University and Department of Computer Science University of Durham, Technical Report EBSE 2007-01, 2007.

[44] M. Harman, S. A. Mansouri, and Y. Zhang, "Search Based Software Engineering: A Comprehensive Analysis and Review of Trends Techniques and Applications," King's College, London, United Kingdom, Technical Report TR-09-03, 2009.

[45] W. Afzal, R. Torkar, and R. Feldt, "A systematic review of search-based testing for non-functional system properties," *Information and Software Technology*, vol. 51, no. 6, pp. 957 – 976, 2009.

[46] M. Hapke, A. Jaszkiewicz, and R. Slowinski, "Fuzzy project scheduling system for software development," *Fuzzy Sets and Systems*, vol. 67, no. 1, pp. 101 – 117, 1994.

[47] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wessln, *Experimentation in Software Engineering*. Springer Publishing Company, Incorporated, 2012.

[48] J. Rumbaugh, I. Jacobson, and G. Booch, *The Unified Modeling Language Reference Manual (2nd Edition)*. Pearson Higher Education, 2004.

[49] D. Kang, J. Jung, and D.-H. Bae, "Constraint-based human resource allocation in software projects," *Software: Practice and Experience*, vol. 41, no. 5, pp. 551–577, 2011.

[50] J. Xiao, L. J. Osterweil, Q. Wang, and M. Li, "Disruption-driven Resource Rescheduling in Software Development Processes," in *Proceedings of the 2010 International Conference on Software Process*, ser. ICSP'10, Paderborn, Germany, 2010, pp. 234–247.

[51] C. Stylianou, S. Gerasimou, and A. Andreou, "A Novel Prototype Tool for Intelligent Software Project Scheduling and Staffing Enhanced with Personality Factors," in *IEEE 24th International Conference on Tools with Artificial Intelligence*, vol. 1, 2012, pp. 277–284.

[52] U. Sanal, "A decision support system for fuzzy scheduling of software projects," in *2000 IEEE AUTOTESTCON Proceedings*, 2000, pp. 263–272.

[53] C. Li, M. Akker, S. Brinkkemper, and G. Diepen, "An integrated approach for requirement selection and scheduling in software release planning," *Requirements Engineering*, vol. 15, no. 4, pp. 375–396, 2010.

[54] C. Robson, *Real World Research*, 2nd ed. Blackwell, 2002.

[55] P. Runeson and M. Höst, "Guidelines for conducting and reporting case study research in software engineering," *Empirical Software Engineering*, vol. 14, pp. 131–164, April 2009.

[56] C. L. B. Maia, T. F. do Nascimento, F. G. Freitas, and J. T. Souza, "An Evolutionary Optimization Approach to Software Test Case Allocation," in *Proceedings of the International Conference on Computational Intelligence and Information Technology*, 2011, pp. 637–641.

[57] R. Britto, P. Neto, R. Rabelo, W. Ayala, and T. Soares, "A hybrid approach to solve the agile team allocation problem," in *2012 IEEE Congress on Evolutionary Computation (CEC)*, 2012, pp. 1–8.

[58] V. Basili, F. Shull, and F. Lanubile, "Building knowledge through families of experiments," *IEEE Transactions on Software Engineering*, vol. 25, no. 4, pp. 456–473, Jul 1999.

[59] D. I. K. Sjøberg, J. E. Hannay, O. Hansen, V. B. Kampenes, A. Karahasanovic, N.-K. Liborg, and A. Rekdal, "A Survey of Controlled Experiments in Software Engineering," *IEEE Transactions on Software Engineering*, vol. 31, no. 9, pp. 733–753, September 2005.

[60] T. Abdel-Hamid and S. E. Madnick, *Software Project Dynamics: An Integrated Approach*. Prentice-Hall, Inc., 1991.

[61] E. O. Navarro, "SimSE: A Software Engineering Simulation Environment for Software Process Education," Ph.D. dissertation, Donald Bren School of Information and Computer Sciences, University of California, Irvine, 2006.

[62] D. C. C. Kupsch, "SPIAL: A Tool for Software Process Improvement Training," Ph.D. dissertation, Federal University of Minas Gerais, Brazil, 2012.

[63] Z. Chen, B. Boehm, T. Menzies, and D. Port, "Finding the Right Data for Software Cost Modeling," *IEEE Software*, vol. 22, no. 6, pp. 38–46, Nov. 2005.

[64] D. I. K. Sjøberg, B. Anda, E. Arisholm, T. Dybå, M. Jørgensen, A. Karahasanovic, E. Koren, and M. Vokác, "Conducting realistic experiments in software engineering," in *Proceedings of the International Symposium on Empirical Software Engineering*, ser. ISESE' 12, 2002, pp. 17–26.

[65] B. Kitchenham, "Procedures for Performing Systematic Reviews," Software Engineering Group, Department of Computer Science, Keele University, Technical Report TR/SE -0401, 2004.